Signals, Systems & the Fourier Transform

A prerequisite revamp for EE341 (Communication Systems–I), IIT Bombay

How to use this. You said your fundamentals are shaky, so this builds each idea up rather than just listing formulas. But your first quiz is **17 August**, so it is ordered by *what EE341 actually uses*, not by what a full signals course would cover.

If you have three days:

- **Day 1** — Part A (§1–§3) and Part B (§4–§6). This is the load-bearing material. Do every worked problem *yourself* first.
- **Day 2** — Part C (§7–§9). Sampling is standard; §9 (bandpass / lowpass-equivalent) is where your lecture notes already are, so it matters most.
- **Day 3** — Part D (§10) skim only, then the problem set in §11 under time pressure.

Notation warning. EE341 (and Madhow's book) uses **ordinary frequency f in Hz**, not ω in rad/s. Your notes write $X(f)$, $\cos(2\pi f_c t)$, never ω. Everything here follows that convention. It removes all the annoying $1/2\pi$ factors — the transform becomes perfectly symmetric.

Contents

PART A — Signals and Systems

1 What a signal is, and the four classifications that matter

A **signal** is a function carrying information. In this course it is almost always $x(t)$, a real-valued function of continuous time. Four classifications come up constantly in communications, and each one is really a question about *which mathematical tool applies.*

1.1 Energy vs. power signals

Define

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 \, \mathrm{d}t, \qquad P_x = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \, \mathrm{d}t.$$

- If $0 < E_x < \infty$, x is an **energy signal**. Then $P_x = 0$ automatically. Pulses, decaying exponentials, anything that dies out.
- If $E_x = \infty$ but $0 < P_x < \infty$, x is a **power signal**. Sinusoids, periodic signals, random noise, and *every real communication waveform* (a transmitter runs forever).

Why you should care. This determines which spectral object exists. An energy signal has a Fourier transform $X(f)$ and an *energy* spectral density $|X(f)|^2$. A power signal generally does *not* have a well-behaved $X(f)$ — it has a *power* spectral density instead. This is exactly why Part D (random processes) exists: modulated signals and noise are power signals, so "take the Fourier transform of the noise" is meaningless, and you must talk about PSD.

Worked problem 1.1. Your notes (page 4) compute the power of the DSB-SC signal $x_c(t) = A\, x(t) \cos(2\pi f_c t)$. Reproduce it.

Solution.

$$P_{x_c} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} A^2 x^2(t) \cos^2(2\pi f_c t) \, \mathrm{d}t = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} A^2 x^2(t) \, \frac{1+\cos(4\pi f_c t)}{2} \, \mathrm{d}t.$$

Split it. The first piece gives $\frac{A^2}{2} P_x$. The second piece, $\frac{A^2}{2} \langle x^2(t) \cos(4\pi f_c t) \rangle$, vanishes: $x^2(t)$ is lowpass with bandwidth $\le 2W$, while $\cos(4\pi f_c t)$ oscillates at $2f_c \gg 2W$, so the product averages to zero. Hence

$$\boxed{P_{x_c} = \frac{A^2}{2} P_x}.$$

For DSB-AM, $x_c'(t) = A(1+\mu x(t)) \cos(2\pi f_c t)$, the same argument on $(1+\mu x)^2 = 1 + 2\mu x + \mu^2 x^2$ with $\langle x \rangle = 0$ gives

$$P_{x_c'} = \frac{A^2}{2}\left(1+\mu^2 P_x\right) = \underbrace{\frac{A^2}{2}}_{\text{carrier, useless}} + \underbrace{\frac{A^2\mu^2}{2} P_x}_{\text{message}}.$$

The **power efficiency** $\eta = \mu^2 P_x / (1 + \mu^2 P_x)$ is why DSB-AM is wasteful — a favourite quiz question.

1.2 Periodic vs. aperiodic

$x(t)$ is periodic if $x(t+T_0) = x(t)$ for some $T_0 > 0$; the smallest such T_0 is the fundamental period, $f_0 = 1/T_0$. Periodic $\Rightarrow$ Fourier *series* (discrete spectrum, spectral lines). Aperiodic $\Rightarrow$ Fourier *transform* (continuous spectrum). §4 shows these are the same theorem.

1.3 Real vs. complex

Physical signals are real. This is not a triviality — it forces **Hermitian symmetry** on the spectrum (§5.4), which is the single most-used structural fact in this course. Complex signals appear as *bookkeeping devices*: the analytic signal $x_+(t)$ and the complex envelope $x_\ell(t)$ in §9 are complex by construction, and that is the whole point.

1.4 Causal vs. non-causal; lowpass vs. bandpass

Causal: $x(t) = 0$ for $t < 0$. Matters for realizability of filters.

Lowpass vs. bandpass is the classification your course opens with. From your notes, page 1:

> $x(t)$ is a **lowpass signal of bandwidth** W if $X(f) = 0$ for all $|f| > W$.
> $x(t)$ is a **bandpass signal** with centre f_c and bandwidth $2W$ if $X(f) = 0$ except for $f_c - W \le |f| \le f_c + W$.

Careful with the word "bandwidth". For a real lowpass signal occupying $[-W, W]$, the bandwidth is W (the one-sided extent), not $2W$. For the bandpass signal occupying $[f_c - W, f_c + W]$ and its mirror, the bandwidth is $2W$. Your notes cross out a "2" on page 1 for exactly this reason. Sloppiness here loses marks; state which convention you are using.

2 Elementary signals you must know cold

Name	Definition	Note
Unit rectangle	$\mathrm{rect}(t) = 1$ for $\lvert t\rvert < \frac{1}{2}$, else 0	width 1, centred at 0
Unit triangle	$\mathrm{tri}(t) = 1 - \lvert t\rvert$ for $\lvert t\rvert < 1$, else 0	$= \mathrm{rect} * \mathrm{rect}$
Sinc	$\mathrm{sinc}(t) = \dfrac{\sin(\pi t)}{\pi t}$, $\mathrm{sinc}(0) = 1$	zeros at all nonzero integers
Unit step	$u(t) = 1$ for $t > 0$, 0 for $t < 0$, $\frac{1}{2}$ at 0	
Signum	$\mathrm{sgn}(t) = 2u(t) - 1$	
Impulse	$\delta(t)$, see below	not a function

> **Trap.** There are two sinc conventions. Communications uses the **normalised** one, $\mathrm{sinc}(t) = \sin(\pi t)/(\pi t)$, so that $\mathrm{rect}(t) \overset{\mathcal{F}}{\longleftrightarrow} \mathrm{sinc}(f)$ cleanly. Some maths and DSP texts use $\sin(t)/t$. Always check. Every formula in this document uses the normalised one.

2.1 The delta function, properly

$\delta(t)$ is not a function; it is defined *entirely* by how it behaves inside an integral:

$$\int_{-\infty}^{\infty} \varphi(t)\,\delta(t - t_0)\,\mathrm{d}t = \varphi(t_0) \qquad \text{(sifting property)}$$

for every continuous φ. Everything else follows from this. Two consequences you will use daily:

$$x(t) * \delta(t - t_0) = x(t - t_0) \qquad \text{(convolving with a shifted impulse = shifting)} \tag{1}$$

$$x(t)\,\delta(t - t_0) = x(t_0)\,\delta(t - t_0) \qquad \text{(multiplying by an impulse = sampling)} \tag{2}$$

Also the scaling rule $\delta(at) = \frac{1}{|a|}\delta(t)$, which trips people up constantly.

Trap. $\delta(t)$ has *unit area*, not unit height. So in a spectrum plot an impulse is drawn as an arrow whose labelled height is its *weight* (area). When your notes draw the carrier line in the DSB-AM spectrum as a tall arrow at f_c, that arrow has weight $A/2$, i.e. the term $\frac{A}{2}\delta(f - f_c)$ — it is not "infinitely large signal".

3 LTI systems and convolution

A system $\mathcal{H}$ maps input $x(t)$ to output $y(t)$. It is
- **Linear** if $\mathcal{H}\{ax_1 + bx_2\} = a\mathcal{H}\{x_1\} + b\mathcal{H}\{x_2\}$;
- **Time-invariant** if $\mathcal{H}\{x(t - t_0)\} = y(t - t_0)$.

Both together: **LTI**.

3.1 Why LTI is the whole game

Here is the argument you should be able to reproduce, because it explains why Fourier analysis exists at all.

Write any input using the sifting property as a continuum of shifted impulses:

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t - \tau)\,\mathrm{d}\tau.$$

Let $h(t) = \mathcal{H}\{\delta(t)\}$ be the **impulse response**. By time-invariance, $\mathcal{H}\{\delta(t - \tau)\} = h(t - \tau)$. By linearity (extended to integrals), push $\mathcal{H}$ through:

$$\boxed{y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t - \tau)\,\mathrm{d}\tau \;=\; (x * h)(t)}$$

So **an LTI system is completely characterised by one function,** $h(t)$. That is an enormous reduction, and it is why we can talk about "the channel" or "the filter" as a single object.

3.2 Properties of convolution

Commutative: $x * h = h * x$	Associative: $(x * h_1) * h_2 = x * (h_1 * h_2)$
Distributive: $x * (h_1 + h_2) = x * h_1 + x * h_2$	Shift: $x(t) * h(t - t_0) = y(t - t_0)$

Associativity is why cascaded filters multiply their frequency responses; distributivity is why parallel branches add.

Widths add. If x is supported on $[a_1, b_1]$ and h on $[a_2, b_2]$, then $x * h$ is supported on $[a_1 + a_2,\, b_1 + b_2]$. Use this as an instant sanity check on any convolution you compute.

Worked problem 3.1. Compute $\mathrm{rect}(t) * \mathrm{rect}(t)$.

Solution. Both have width 1, so the answer has width 2, supported on $[-1, 1]$ — already a check.

$$(\mathrm{rect} * \mathrm{rect})(t) = \int_{-\infty}^{\infty} \mathrm{rect}(\tau)\,\mathrm{rect}(t - \tau)\,\mathrm{d}\tau = \int_{-1/2}^{1/2} \mathrm{rect}(t - \tau)\,\mathrm{d}\tau.$$

The integrand is 1 when $|t - \tau| < \frac{1}{2}$, i.e. $\tau \in (t - \frac{1}{2},\, t + \frac{1}{2})$. So the integral is the length of the overlap between $(-\frac{1}{2}, \frac{1}{2})$ and $(t - \frac{1}{2}, t + \frac{1}{2})$, which is $1 - |t|$ for $|t| \le 1$ and 0 otherwise. Therefore

$$\mathrm{rect}(t) * \mathrm{rect}(t) = \mathrm{tri}(t).$$

This one result will save you repeatedly: it is why sinc^2 pairs with the triangle, and it is the reason a raised-cosine-type pulse shows up later in the ISI chapter.

3.3 Causality and stability from $h(t)$

- **Causal** $\iff$ $h(t) = 0$ for $t < 0$.
- **BIBO stable** $\iff$ $\int |h(t)|\,\mathrm{d}t < \infty$.

The ideal lowpass filter has $h(t) = 2W\,\mathrm{sinc}(2Wt)$, which is neither causal nor absolutely integrable. That is why "ideal filter" is a mathematical fiction we use for analysis and approximate in practice — worth one sentence in any exam answer that invokes one.

3.4 Eigenfunctions: the bridge to Fourier

Feed a complex exponential $x(t) = e^{\mathrm{j}2\pi f_0 t}$ into an LTI system:

$$y(t) = \int h(\tau)e^{\mathrm{j}2\pi f_0(t-\tau)}\,\mathrm{d}\tau = e^{\mathrm{j}2\pi f_0 t}\underbrace{\int h(\tau)e^{-\mathrm{j}2\pi f_0 \tau}\,\mathrm{d}\tau}_{H(f_0)} = H(f_0)\,e^{\mathrm{j}2\pi f_0 t}.$$

The complex exponential comes out unchanged except for a complex scale factor. It is an *eigenfunction* of every LTI system, with eigenvalue $H(f_0)$ — and that eigenvalue is precisely the Fourier transform of h.

This is *the* reason we decompose signals into complex exponentials rather than, say, polynomials: because it turns the hard operation (convolution) into the easy one (multiplication). Everything in Part B is machinery for doing that decomposition.

Trap. $\cos(2\pi f_0 t)$ is *not* an eigenfunction — an LTI system can change its phase, so the output is $|H(f_0)|\cos(2\pi f_0 t + \angle H(f_0))$, not a scalar multiple of the input. Only the complex exponential is a true eigenfunction. (For a real system the cosine result follows by writing it as a sum of two exponentials and using $H(-f) = H^*(f)$.)

PART B — The Fourier Transform

4 From Fourier series to Fourier transform

4.1 The series

Any reasonable periodic signal with period T_0 ($f_0 = 1/T_0$) can be written as

$$x(t) = \sum_{n=-\infty}^{\infty} c_n e^{\mathrm{j}2\pi n f_0 t}, \qquad c_n = \frac{1}{T_0}\int_{T_0} x(t)e^{-\mathrm{j}2\pi n f_0 t}\,\mathrm{d}t.$$

The c_n are complex numbers: $|c_n|$ is the amplitude of the n-th harmonic, $\angle c_n$ its phase. The spectrum is a set of **discrete lines** at $f = nf_0$.

Why does the formula for c_n work? Because the exponentials are orthogonal over a period: $\frac{1}{T_0}\int_{T_0} e^{\mathrm{j}2\pi(n-m)f_0 t}\,\mathrm{d}t = \delta_{nm}$. Multiplying the series by $e^{-\mathrm{j}2\pi m f_0 t}$ and averaging kills every term except $n = m$. That is the entire trick, and the same trick reappears in matched filtering and in orthogonal signalling later in the course.

Worked problem 4.1. Find the Fourier series of the square wave $c(t)$ your notes use for the ring modulator / switching demodulator: $c(t) = +1$ for the first half of each period, -1 for the second, period $T_c = 1/f_c$.

Solution. $c(t)$ is odd (with the origin chosen at a rising edge midpoint) and has zero mean, so only odd harmonics survive:

$$c(t) = \frac{4}{\pi}\left[\cos(2\pi f_c t) - \frac{1}{3}\cos(6\pi f_c t) + \frac{1}{5}\cos(10\pi f_c t) - \cdots\right].$$

That matches your notes exactly ($\frac{4}{\pi}\cos\omega_c t - \frac{4}{3\pi}\cos 3\omega_c t + \frac{4}{5\pi}\cos 5\omega_c t$). The *unipolar* switching function (0/1 instead of ∓ 1) that appears on the demodulation page is just $\frac{1+c(t)}{2}$:

$$s(t) = \frac{1}{2} + \frac{2}{\pi}\cos(2\pi f_c t) - \frac{2}{3\pi}\cos(6\pi f_c t) + \cdots$$

— again exactly your notes. **Why this matters:** a switching modulator multiplies $x(t)$ by $s(t)$, so the output contains $x(t)$ replicated around DC, f_c, $3f_c$, $5f_c, \ldots$ A bandpass filter at f_c with bandwidth $2W$ picks off the term $\frac{2}{\pi}x(t)\cos(2\pi f_c t)$, which is DSB-SC. Cheap hardware, exact result.

4.2 Letting the period go to infinity

Take an aperiodic $x(t)$, and build a periodic version $\tilde{x}_{T_0}$ by repeating it every T_0. Then

$$c_n = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)e^{-\mathrm{j}2\pi n f_0 t}\,\mathrm{d}t = \frac{1}{T_0}X(nf_0), \qquad X(f) := \int_{-\infty}^{\infty} x(t)e^{-\mathrm{j}2\pi f t}\,\mathrm{d}t.$$

The lines sit at spacing $f_0 = 1/T_0$ and have height $X(nf_0)/T_0$. As $T_0 \to \infty$ the lines merge into a continuum, the sum becomes an integral, and we get the transform pair.

The Fourier transform pair (Hz convention).

$$X(f) = \int_{-\infty}^{\infty} x(t)\,e^{-\mathrm{j}2\pi f t}\,\mathrm{d}t, \qquad x(t) = \int_{-\infty}^{\infty} X(f)\,e^{+\mathrm{j}2\pi f t}\,\mathrm{d}f.$$

Note the beautiful symmetry: no $1/2\pi$ anywhere, and forward/inverse differ only in the sign of the exponent. This is the payoff for using f instead of ω.

Converting from an ω-based textbook. If a book gives $X(\omega)$, then $X(f) = X(\omega)\big|_{\omega=2\pi f}$ — the *function values* are the same, only the argument is relabelled. But a formula like $x(t) = \frac{1}{2\pi}\int X(\omega)e^{\mathrm{j}\omega t}\,\mathrm{d}\omega$ carries a $1/2\pi$ that disappears in the f convention. Never mix the two mid-problem.

4.3 What $X(f)$ actually means

$X(f)$ is complex. Write $X(f) = |X(f)|e^{\mathrm{j}\theta(f)}$.

- $|X(f)|$ — **magnitude spectrum**: how much of frequency f is present.
- $\theta(f)$ — **phase spectrum**: *where* those components sit in time. Phase carries most of the perceptual information in a signal; discarding it destroys the waveform even though $|X|$ is intact.
- Units: if x is in volts, $X(f)$ is in volts/Hz. It is a *density*, which is why impulses appear whenever a signal has finite power concentrated at a single frequency.

5 Properties of the Fourier transform

These are not trivia to memorise — in this course you will almost never compute an FT integral. You will recognise a signal as a combination of known pairs and apply properties. Learn them as *moves*.

5.1 Linearity

$ax_1(t) + bx_2(t) \xleftrightarrow{\mathcal{F}} aX_1(f) + bX_2(f)$. Obvious, but it is what lets you decompose DSB-AM into carrier + sidebands and treat each separately.

5.2 Time shift and frequency shift (the two you will use most)

$$x(t-t_0) \overset{\mathcal{F}}{\longleftrightarrow} X(f)\,e^{-j2\pi f t_0} \qquad \text{delay} \Rightarrow \text{linear phase, magnitude unchanged} \tag{3}$$

$$x(t)\,e^{j2\pi f_c t} \overset{\mathcal{F}}{\longleftrightarrow} X(f-f_c) \qquad \text{multiply by exponential} \Rightarrow \text{shift spectrum} \tag{4}$$

Proof of the second (do this once by hand, it takes two lines): $\int x(t)e^{j2\pi f_c t}e^{-j2\pi f t}\,dt = \int x(t)e^{-j2\pi(f-f_c)t}\,dt = X(f-f_c)$.

The modulation property — the single most important formula in this course.

$$x(t)\cos(2\pi f_c t) \;=\; \frac{x(t)e^{j2\pi f_c t} + x(t)e^{-j2\pi f_c t}}{2} \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}X(f-f_c) + \frac{1}{2}X(f+f_c).$$

Multiplying by a carrier **takes the spectrum, halves it, and puts a copy at $+f_c$ and a copy at $-f_c$**. That is literally the definition of modulation as your notes state it: "process of shifting frequency". Everything in the AM chapter is this one line applied repeatedly.

Your notes write $X_c(f) = \frac{A}{2}X(f-f_c) + \frac{A}{2}X(f+f_c)$ for $x_c(t) = Ax(t)\cos(2\pi f_c t)$, and then say "$= \frac{A}{2}X(f-f_c)$ for $f > 0$, and $X_c(f)$ has Hermitian symmetry". That shorthand is legitimate *only* because $X(f-f_c)$ and $X(f+f_c)$ do not overlap, which needs $f_c > W$. State that condition.

5.3 Scaling

$$x(at) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{|a|}X\left(\frac{f}{a}\right).$$

Compress in time $\Rightarrow$ stretch in frequency. This is the time–bandwidth trade-off: a short pulse necessarily occupies a wide band. It is the reason high data rates need wide bandwidth, which is the theme of the second half of EE341.

5.4 Duality

If $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(f)$, then $X(t) \overset{\mathcal{F}}{\longleftrightarrow} x(-f)$. This gives you a free pair for every pair you know: from $\mathrm{rect}(t) \overset{\mathcal{F}}{\longleftrightarrow} \mathrm{sinc}(f)$ you immediately get $\mathrm{sinc}(t) \overset{\mathcal{F}}{\longleftrightarrow} \mathrm{rect}(f)$ (using evenness). Use it; it halves what you need to memorise.

5.5 Conjugate symmetry — Hermitian symmetry

If $x(t)$ is **real**, then

$$\boxed{X(-f) = X^*(f)}$$

equivalently $|X(f)|$ is **even** and $\angle X(f)$ is **odd**.

Proof. $X(-f) = \int x(t)e^{j2\pi f t}\,dt = \left[\int x^*(t)e^{-j2\pi f t}\,dt\right]^* = X^*(f)$ since $x = x^*$.

Why this dominates the course. It says the negative-frequency half of the spectrum of a real signal carries *no new information*. Hence:

- You may specify a real bandpass signal by its positive-frequency content alone — this is what your notes mean by "$X_c(f) = AX(f-f_c)$ for $f > 0$, and $X_c(f)$ has Hermitian symmetry".
- It motivates the **analytic signal** $x_+(t)$ (§9), which simply *deletes* the redundant half.
- It is why DSB transmission is spectrally wasteful (upper and lower sidebands are mirror images), which motivates SSB and VSB.

Two corollaries worth having reflexively: x real and even $\Rightarrow$ X real and even; x real and odd $\Rightarrow$ X purely imaginary and odd.

5.6 Convolution and multiplication

$$x(t) * h(t) \overset{\mathcal{F}}{\longleftrightarrow} X(f)H(f) \qquad\qquad x(t)\,h(t) \overset{\mathcal{F}}{\longleftrightarrow} X(f) * H(f)$$

The first is the reason Fourier analysis is used at all (§3.5). The second is its dual and is how you handle any product of signals — e.g. the square-law modulator, where you square $x(t) + \cos 2\pi f_c t$ and want to know where the resulting spectral pieces land.

Proof of the first (know it):

$$\int\left[\int x(\tau)h(t-\tau)\,\mathrm{d}\tau\right]e^{-\mathrm{j}2\pi ft}\,\mathrm{d}t = \int x(\tau)\left[\int h(t-\tau)e^{-\mathrm{j}2\pi ft}\,\mathrm{d}t\right]\mathrm{d}\tau = \int x(\tau)H(f)e^{-\mathrm{j}2\pi f\tau}\,\mathrm{d}\tau = X(f)H(f),$$

where the inner bracket used the time-shift property.

5.7 Differentiation and integration

$$\frac{\mathrm{d}x}{\mathrm{d}t} \overset{\mathcal{F}}{\longleftrightarrow} \mathrm{j}2\pi f\,X(f), \qquad \int_{-\infty}^{t} x(\tau)\,\mathrm{d}\tau \overset{\mathcal{F}}{\longleftrightarrow} \frac{X(f)}{\mathrm{j}2\pi f} + \frac{X(0)}{2}\delta(f).$$

The $\delta(f)$ term (DC offset) is forgotten by almost everyone. It is needed exactly when $X(0) \neq 0$.

5.8 Parseval / Rayleigh

$$\int_{-\infty}^{\infty} |x(t)|^2\,\mathrm{d}t = \int_{-\infty}^{\infty} |X(f)|^2\,\mathrm{d}f \qquad \text{(energy signals)}$$

More generally $\int x(t)y^*(t)\,\mathrm{d}t = \int X(f)Y^*(f)\,\mathrm{d}f$. The quantity $|X(f)|^2$ is the **energy spectral density** in J/Hz. For *power* signals the analogue is the power spectral density of §10 — do not apply this Parseval to a modulated carrier.

5.9 Moment / area rules (fast sanity checks)

$$X(0) = \int x(t)\,\mathrm{d}t, \qquad x(0) = \int X(f)\,\mathrm{d}f.$$

Use these constantly. If you compute a transform and $X(0)$ does not equal the area under $x(t)$, you have made an algebra error. Thirty seconds, catches most mistakes.

6 Standard transform pairs

$x(t)$	$X(f)$
$\delta(t)$	1
1	$\delta(f)$
$\delta(t-t_0)$	$e^{-\mathrm{j}2\pi f t_0}$
$e^{\mathrm{j}2\pi f_c t}$	$\delta(f-f_c)$
$\cos(2\pi f_c t)$	$\frac{1}{2}\delta(f-f_c)+\frac{1}{2}\delta(f+f_c)$
$\sin(2\pi f_c t)$	$\frac{1}{2\mathrm{j}}\delta(f-f_c)-\frac{1}{2\mathrm{j}}\delta(f+f_c)$
$\mathrm{rect}(t/T)$	$T\,\mathrm{sinc}(fT)$
$2W\,\mathrm{sinc}(2Wt)$	$\mathrm{rect}\big(f/(2W)\big)$
$\mathrm{tri}(t/T)$	$T\,\mathrm{sinc}^2(fT)$
$e^{-at}u(t)$, $a>0$	$\dfrac{1}{a+\mathrm{j}2\pi f}$
$e^{-a\lvert t\rvert}$, $a>0$	$\dfrac{2a}{a^2+(2\pi f)^2}$
$e^{-\pi t^2}$	$e^{-\pi f^2}$ (its own transform)
$u(t)$	$\dfrac{1}{2}\delta(f)+\dfrac{1}{\mathrm{j}2\pi f}$
$\mathrm{sgn}(t)$	$\dfrac{1}{\mathrm{j}\pi f}$
$\dfrac{1}{\pi t}$	$-\mathrm{j}\ \mathrm{sgn}(f)$ **(Hilbert transform kernel)**
$\displaystyle\sum_n \delta(t-nT_s)$	$\displaystyle\frac{1}{T_s}\sum_k \delta\Big(f-\frac{k}{T_s}\Big)$ **(impulse train)**

The three lines to memorise above all others, because EE341 uses them on nearly every page:

1. $\cos(2\pi f_c t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}[\delta(f-f_c)+\delta(f+f_c)]$ — modulation.
2. $\dfrac{1}{\pi t} \overset{\mathcal{F}}{\longleftrightarrow} -\mathrm{j}\,\mathrm{sgn}(f)$ — Hilbert transform, hence analytic signal and SSB.
3. $\sum_n \delta(t-nT_s) \overset{\mathcal{F}}{\longleftrightarrow} \dfrac{1}{T_s}\sum_k \delta(f-k/T_s)$ — sampling theorem.

Worked problem 6.1. Derive $u(t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}\delta(f)+\frac{1}{\mathrm{j}2\pi f}$, and use it to get your notes' expression $U(f)=\frac{1}{2}(1+\mathrm{sgn}(f))$ in the frequency variable.

Solution. Write $u(t)=\frac{1}{2}+\frac{1}{2}\,\mathrm{sgn}(t)$. The constant $\frac{1}{2}$ transforms to $\frac{1}{2}\delta(f)$. For $\mathrm{sgn}(t)$: it is not integrable, so take the limit of $e^{-a|t|}\,\mathrm{sgn}(t)$ as $a\to 0^+$:

$$\int_0^\infty e^{-at}e^{-\mathrm{j}2\pi ft}\,\mathrm{d}t-\int_{-\infty}^0 e^{at}e^{-\mathrm{j}2\pi ft}\,\mathrm{d}t=\frac{1}{a+\mathrm{j}2\pi f}-\frac{1}{a-\mathrm{j}2\pi f}\xrightarrow{a\to 0^+}\frac{-\mathrm{j}4\pi f}{(2\pi f)^2}=\frac{1}{\mathrm{j}\pi f}.$$

Hence $u(t) \overset{\mathcal{F}}{\longleftrightarrow} \frac{1}{2}\delta(f) + \frac{1}{\mathrm{j}2\pi f}$.

Now the notes' $U(f)$ is a *different object*: it is the unit step *as a function of frequency*, used as a mask to kill negative frequencies. There, $U(f) = \frac{1}{2}(1 + \operatorname{sgn}(f))$ is simply the algebraic identity $u = \frac{1}{2} + \frac{1}{2}\operatorname{sgn}$ written with argument f: $U(f) = 0$ for $f < 0$, $\frac{1}{2}$ at $f = 0$, 1 for $f > 0$ — precisely the box on your SNR page. Keep the two straight: one is a transform, the other is a masking function.

Worked problem 6.2. A pulse $p(t) = \operatorname{rect}(t/T)$ is transmitted. Find its bandwidth by the first-null definition, and comment on the time–bandwidth product.

Solution. $P(f) = T\operatorname{sinc}(fT)$, first null at $f = 1/T$. So $B_{\text{null}} = 1/T$ and $B\cdot T = 1$. Halving the pulse duration doubles the bandwidth — the scaling property in action. This constant is the seed of the Nyquist criterion in the ISI chapter: you cannot signal faster than roughly $2B$ symbols/second in bandwidth B.

PART C — Filtering, Sampling, and Bandpass Signals

7 Filtering in the frequency domain

For an LTI system, $Y(f) = H(f)X(f)$ where $H(f)$ is the **frequency response** (the FT of $h(t)$). Magnitude $|H(f)|$ scales each frequency; phase $\angle H(f)$ delays it.

7.1 Distortionless transmission

A channel is distortionless if $y(t) = K\,x(t - t_d)$, i.e.

$$H(f) = Ke^{-\mathrm{j}2\pi f t_d} \quad\Longleftrightarrow\quad |H(f)| = K \text{ (flat)}, \qquad \angle H(f) = -2\pi f t_d \text{ (linear in } f).$$

Flat magnitude and *linear* phase over the signal's band. Nonflat magnitude gives amplitude distortion; nonlinear phase gives **delay distortion** — different frequency components arrive at different times, smearing pulses. This is the same phenomenon that becomes intersymbol interference later. The *group delay* $\tau_g(f) = -\frac{1}{2\pi}\frac{\mathrm{d}\angle H(f)}{\mathrm{d}f}$ is constant exactly when phase is linear.

7.2 Ideal filters

- **Ideal LPF**, cutoff W: $H(f) = \operatorname{rect}\big(\frac{f}{2W}\big)$, so $h(t) = 2W\operatorname{sinc}(2Wt)$.
- **Ideal BPF**, centre f_c, bandwidth $2W$: $H(f) = \operatorname{rect}\big(\frac{f-f_c}{2W}\big) + \operatorname{rect}\big(\frac{f+f_c}{2W}\big)$, so $h(t) = 4W\operatorname{sinc}(2Wt)\cos(2\pi f_c t)$ — a lowpass sinc *modulated* onto the carrier. Note how the modulation property gives you the bandpass impulse response for free.

Both are non-causal (the sinc extends to $t < 0$) and unstable in the BIBO sense. Real filters (Butterworth, Chebyshev) approximate them with finite roll-off, and a real receiver's "high-Q tunable bandpass filter" — the thing your superheterodyne page says is hard to build — is exactly this approximation problem.

Worked problem 7.1. In the coherent demodulator on your notes page 7: $x_c'(t) = Ax(t)\cos(2\pi f_c t)$ is multiplied by $\cos(2\pi f_c t + \phi)$ and lowpass filtered. Find the output.

Solution.

$$y(t) = Ax(t)\cos(2\pi f_c t)\cos(2\pi f_c t + \phi) = \frac{Ax(t)}{2}\big[\cos\phi + \cos(4\pi f_c t + \phi)\big].$$

The LPF (cutoff $W < 2f_c - W$) removes the $2f_c$ term, leaving

$$\boxed{y_{\text{out}}(t) = \frac{A\cos\phi}{2}\,x(t)}.$$

The punchline of coherent detection. The recovered signal is scaled by $\cos\phi$. If the local oscillator is 90° out of phase ($\phi = \pi/2$) the output is *zero* — total loss of signal. This is the **quadrature null effect**, and it is why DSB-SC needs carrier recovery (a PLL), and why DSB-AM with envelope detection — which needs no phase reference at all — survived commercially despite being power-inefficient.

If instead there is a frequency error Δf (LO at $f_c+\Delta f$), you get $\frac{A}{2}x(t)\cos(2\pi\Delta f\,t)$: a slow beating in and out. Your notes' page 11 does exactly this calculation in complex-envelope form.

8 Sampling

8.1 Impulse sampling

Sample $x(t)$ every T_s seconds. Model it as multiplication by an impulse train:

$$x_s(t) = x(t)\sum_n \delta(t - nT_s) = \sum_n x(nT_s)\delta(t - nT_s).$$

Multiplication in time ⇒ convolution in frequency, and the impulse train transforms to another impulse train with spacing $f_s = 1/T_s$:

$$X_s(f) = X(f) * \frac{1}{T_s}\sum_k \delta(f - kf_s) = \boxed{f_s \sum_{k=-\infty}^{\infty} X(f - kf_s)}$$

Sampling replicates the spectrum at every multiple of f_s, scaled by f_s. That single sentence is the sampling theorem; everything else is bookkeeping about whether the replicas collide.

8.2 The Nyquist criterion and aliasing

If x is lowpass with bandwidth W (so $X(f) = 0$ for $|f| > W$), the replicas occupy $[kf_s-W,\ kf_s+W]$. They do not overlap iff

$$f_s > 2W \qquad (\text{Nyquist rate} = 2W).$$

Then $x(t)$ is recovered exactly by an ideal LPF of gain T_s and cutoff between W and $f_s - W$:

$$x(t) = \sum_n x(nT_s)\,\operatorname{sinc}\!\left(\frac{t - nT_s}{T_s}\right) \qquad (\text{at } f_s = 2W).$$

If $f_s < 2W$, replicas overlap and high frequencies fold down into the baseband as lower frequencies. This is **aliasing**, and it is *irreversible* — no filter afterwards can undo it. Hence every practical sampler is preceded by an **anti-aliasing filter**.

The equality case. At exactly $f_s = 2W$ the replicas touch at $f = W$. This is fine for a strictly bandlimited signal with no impulse at $\pm W$, but fails for e.g. $x(t) = \sin(2\pi Wt)$ sampled at $2W$: every sample lands on a zero crossing and you recover nothing. Exam questions love this. Say "$f_s > 2W$, with equality permissible only if $X(f)$ has no impulse at $\pm W$".

8.3 Practical sampling

- **Natural sampling**: multiply by a pulse train of finite-width rectangles. Spectrum is $\sum_k c_k X(f - kf_s)$ with c_k the pulse train's Fourier coefficients — replicas are weighted by a sinc envelope but each is undistorted, so recovery is still exact.
- **Flat-top sampling (PAM, sample-and-hold)**: each sample is held for width τ. Now $X_s(f) = \tau\,\mathrm{sinc}(f\tau) \cdot f_s \sum_k X(f - kf_s)$ — the baseband copy is *multiplied* by $\mathrm{sinc}(f\tau)$, which droops across the band. This is **aperture distortion**, fixed by an equaliser of response $1/\mathrm{sinc}(f\tau)$. This appears in EE341's PAM/PCM section.

Worked problem 8.1. $x(t) = \cos(2\pi \cdot 300t)$ is sampled at $f_s = 500$ Hz. What is heard after ideal lowpass reconstruction at 250 Hz?

Solution. $W = 300 > f_s/2 = 250$, so aliasing. Replicas of the lines at ± 300 appear at $\pm 300 + k \cdot 500$. Taking $k = -1$: $300 - 500 = -200$; taking $k = +1$: $-300 + 500 = 200$. So a pair of lines lands at ± 200 Hz inside the reconstruction band. Output: a 200 Hz tone. The alias frequency is $|f - kf_s|$ folded into $[0, f_s/2]$.

9 Bandpass signals and the lowpass equivalent

This is where your lecture notes currently are, and it is the conceptual heart of the first half of EE341. Everything below is a direct consequence of Hermitian symmetry (§5.4).

9.1 The motivation

A real bandpass signal at $f_c = 900$ MHz with bandwidth 200 kHz is a nightmare to simulate or analyse directly — you would need to sample at ≥ 1.8 GHz to represent a signal that carries only 200 kHz of information. The redundancy is twofold: (i) negative frequencies duplicate positive ones (Hermitian symmetry), and (ii) the carrier itself carries no information. Strip both away and you get a *lowpass* complex signal of bandwidth W that contains everything. That object is the complex envelope.

9.2 Step 1: the analytic signal

Delete negative frequencies:

$$X_+(f) = X_{bp}(f)\,U(f), \qquad U(f) = \tfrac{1}{2}\big(1 + \mathrm{sgn}(f)\big).$$

So $X_+(f) = \frac{1}{2}X_{bp}(f) + \frac{1}{2}X_{bp}(f)\,\mathrm{sgn}(f)$. In the time domain, using $\mathrm{sgn}(f) \xleftrightarrow{\mathcal{F}}{}^{-1}$-pairing with $\frac{1}{\mathrm{j}\pi t}$, i.e. $\frac{1}{\pi t} \xleftrightarrow{\mathcal{F}} -\mathrm{j}\,\mathrm{sgn}(f)$:

$$x_+(t) = \frac{1}{2}x_{bp}(t) + \frac{\mathrm{j}}{2}\,\hat{x}_{bp}(t), \qquad \hat{x}_{bp}(t) := x_{bp}(t) * \frac{1}{\pi t}$$

where $\hat{x}$ is the **Hilbert transform** of x. (Your notes write $x_+ = x_{bp} * \big(\frac{1}{2}\delta(t) + \frac{1}{\mathrm{j}2\pi t}\big)$ — same thing; the $\frac{1}{\mathrm{j}2\pi t}$ combines with the factor to give $\frac{\mathrm{j}}{2}\hat{x}_{bp}$. Watch the j's.)

The Hilbert transform is a $-90°$ phase shifter. Its frequency response is $H(f) = -\mathrm{j}\,\mathrm{sgn}(f)$: magnitude 1 everywhere, phase $-90°$ for $f > 0$ and $+90°$ for $f < 0$. It does nothing to amplitudes; it only rotates phase. Key facts:

$$\widehat{\cos(2\pi f_c t)} = \sin(2\pi f_c t), \qquad \widehat{\sin(2\pi f_c t)} = -\cos(2\pi f_c t), \qquad \hat{\hat{x}} = -x.$$

9.3 Step 2: shift down to baseband

$$x_\ell(t) = 2x_+(t)e^{-j2\pi f_c t} \qquad \text{(some texts, and your notes, omit the 2; be consistent)}$$

Expanding with $x_+ = \frac{1}{2}(x_{bp} + j\hat{x}_{bp})$ and $e^{-j2\pi f_c t} = \cos - j\sin$:

$$x_\ell(t) = x_I(t) + j\, x_Q(t)$$

with

$$x_I(t) = x_{bp}(t)\cos(2\pi f_c t) + \hat{x}_{bp}(t)\sin(2\pi f_c t) \qquad \textbf{in-phase} \tag{5}$$
$$x_Q(t) = \hat{x}_{bp}(t)\cos(2\pi f_c t) - x_{bp}(t)\sin(2\pi f_c t) \qquad \textbf{quadrature} \tag{6}$$

Both x_I and x_Q are **real lowpass signals of bandwidth W**. This is the block diagram on your notes page 12.

9.4 Step 3: reconstruct

$$\boxed{x_{bp}(t) = \mathrm{Re}\left\{x_\ell(t)\, e^{j2\pi f_c t}\right\} = x_I(t)\cos(2\pi f_c t) - x_Q(t)\sin(2\pi f_c t)}$$

That minus sign is not optional — it comes from $\mathrm{Re}\{(x_I + jx_Q)(\cos + j\sin)\}$. Losing it is the most common error in this topic.

9.5 Envelope and phase

Write $x_\ell(t) = A(t)e^{j\phi(t)}$ with

$$A(t) = \sqrt{x_I^2(t) + x_Q^2(t)} \ \ (\geq 0), \qquad \phi(t) = \tan^{-1}\frac{x_Q(t)}{x_I(t)}.$$

Then $x_{bp}(t) = A(t)\cos\big(2\pi f_c t + \phi(t)\big)$. $A(t)$ is the **envelope** — and it is exactly what an envelope detector (diode + RC) outputs.

Now the AM results become one-liners.

- **DSB-SC:** $x_c(t) = Ax(t)\cos 2\pi f_c t = \mathrm{Re}\{Ax(t)e^{j2\pi f_c t}\}$, so $x_\ell = Ax(t)$, which is real. Envelope $= A|x(t)|$. Since $x(t)$ goes negative, the envelope is the *rectified* message — so an envelope detector **fails**, and the sign flips show up as the 180° phase reversals drawn on your notes page 3.
- **DSB-AM:** $x_c'(t) = A(1 + \mu x(t))\cos 2\pi f_c t$, so $x_\ell = A(1 + \mu x(t))$. If $\mu|x(t)| < 1$ for all t (i.e. modulation index under 100%), then $x_\ell > 0$ always, the envelope is $A(1 + \mu x(t))$, and an envelope detector recovers $x(t)$ up to a DC offset. **Over-modulation** ($\mu|x| > 1$) makes x_ℓ change sign and the envelope detector produces the rectified, distorted mess.
- **Angle modulation:** $A(t) = \text{const}$, all the information is in $\phi(t)$.

9.6 Noise in the same language

Your notes page 11 does this: pass bandpass noise $n(t)$ through the receiver's BPF and write $n(t) = n_I(t)\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$. Then received signal + noise has envelope

$$A(t) = \sqrt{\big(x_I(t) + n_I(t)\big)^2 + \big(x_Q(t) + n_Q(t)\big)^2},$$

which is the starting point for every SNR calculation in analog communication. For white Gaussian noise filtered to bandwidth $2W$, n_I and n_Q turn out to be independent, zero-mean, each with the *same* variance as n — proved with the tools of §10.

Worked problem 9.1. Find the lowpass equivalent of $x_{bp}(t) = \cos\big(2\pi(f_c + \Delta f)t\big)$ with respect to carrier f_c.

Solution. $x_{bp} = \mathrm{Re}\{e^{j2\pi(f_c+\Delta f)t}\} = \mathrm{Re}\{e^{j2\pi\Delta f t}e^{j2\pi f_c t}\}$, so $x_\ell(t) = e^{j2\pi\Delta f t}$, giving $x_I = \cos(2\pi\Delta f t)$, $x_Q = \sin(2\pi\Delta f t)$, envelope $A(t) = 1$. **A frequency offset becomes a slow rotation of the complex envelope** — which is precisely the picture on your notes page 11 where $x_\ell(t) = Ax(t) + Ae^{j2\pi\Delta f t}$ and hence $x_I = Ax(t) + A\cos(2\pi\Delta f t)$, $x_Q = A\sin(2\pi\Delta f t)$. Everything about carrier offset, phase noise, and synchronisation is easier to see in this representation.

Worked problem 9.2. Show the bandpass filter $h_{bp}(t) = 2\,\mathrm{Re}\{h_\ell(t)e^{j2\pi f_c t}\}$ acting on x_{bp} produces output whose complex envelope is $y_\ell = x_\ell * h_\ell$.

Solution. Sketch: in the frequency domain, $Y_{bp}(f) = H_{bp}(f)X_{bp}(f)$. Restrict to $f > 0$, where $X_{bp}(f) = \frac{1}{2}X_\ell(f - f_c)$ and $H_{bp}(f) = H_\ell(f - f_c)$. Multiply, shift back by f_c, and read off $Y_\ell(f) = X_\ell(f)H_\ell(f)$; invert. **Why this matters:** it means you can simulate an entire band-pass communication link — modulator, channel filter, receiver filter, noise — entirely at baseband with complex signals, at a sampling rate set by W and not by f_c. That is how every real simulator (and every software-defined radio) works.

PART D — Random Processes Primer

10 Just enough randomness to survive the noise chapter

Skim this before the quiz; return to it properly before the mid-semester. The syllabus lists it as a prereq because noise is not a signal you can write down — you can only describe its statistics.

10.1 Random variables, in one box

Mean $\mu_X = E[X]$; variance $\sigma_X^2 = E[(X-\mu_X)^2] = E[X^2] - \mu_X^2$.
Gaussian: $p_X(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/2\sigma^2}$.
The Q-function: $Q(x) = P(X > x)$ for standard Gaussian $= \displaystyle\int_x^\infty \frac{1}{\sqrt{2\pi}}e^{-t^2/2}\,dt$.

$Q(\cdot)$ is the function every digital-communication error probability is expressed in. Get comfortable with $Q(x) = 1 - Q(-x)$ and its rapid decay. You will meet $P_e = Q\Big(\sqrt{2E_b/N_0}\Big)$ for BPSK.

10.2 Random processes

A random process $X(t)$ is a family of random variables, one per time instant. Two statistics matter:

$$\mu_X(t) = E[X(t)], \qquad R_X(t_1,t_2) = E[X(t_1)X(t_2)] \quad \text{(autocorrelation)}.$$

Wide-sense stationary (WSS) means: $\mu_X(t) = \mu_X$ is constant, and $R_X(t_1,t_2)$ depends only on the lag $\tau = t_1 - t_2$, so we write $R_X(\tau)$. Almost every noise model in this course is WSS.

Properties of $R_X(\tau)$ for a WSS process:

- $R_X(0) = E[X^2(t)] =$ average power.
- $R_X(-\tau) = R_X(\tau)$ (even).
- $|R_X(\tau)| \le R_X(0)$.

10.3 Power spectral density and the Wiener–Khinchin theorem

$$\boxed{S_X(f) = \int_{-\infty}^{\infty} R_X(\tau) e^{-j2\pi f\tau}\,\mathrm{d}\tau} \qquad \text{and} \qquad P_X = R_X(0) = \int_{-\infty}^{\infty} S_X(f)\,\mathrm{d}f.$$

The PSD is the Fourier transform of the autocorrelation. This is what replaces $|X(f)|^2$ when the signal is a power signal with no ordinary Fourier transform.

$S_X(f)$ is real, even, and non-negative. It has units W/Hz.

10.4 The one result you will use constantly

WSS process through an LTI filter. If $Y = X * h$, then

$$S_Y(f) = |H(f)|^2 S_X(f), \qquad \mu_Y = \mu_X H(0).$$

Note $|H(f)|^2$, *not* $H(f)$: phase information is destroyed, because the PSD does not know about phase.

10.5 White noise

AWGN (additive white Gaussian noise) has

$$S_N(f) = \frac{N_0}{2} \text{ (W/Hz, two-sided, for all } f\text{)}, \qquad R_N(\tau) = \frac{N_0}{2}\delta(\tau).$$

Two consequences: (i) samples at distinct times are uncorrelated (and, being Gaussian, independent); (ii) it has infinite power, so it is a fiction — but a harmless one, because every receiver filters it. After an ideal BPF of bandwidth $2W$ (both sidebands, at $\pm f_c$), the noise power is

$$P_N = \int |H(f)|^2 \frac{N_0}{2}\,\mathrm{d}f = \frac{N_0}{2} \cdot 2 \cdot (2W) = 2N_0 W.$$

Combine this with the signal power from Worked Problem 1.1 and you have the SNR expressions your notes are building towards.

Worked problem 10.1. Show that for bandpass noise $n(t) = n_I(t)\cos 2\pi f_c t - n_Q(t)\sin 2\pi f_c t$ with $S_N(f)$ flat of height $N_0/2$ over $|f \mp f_c| < W$, the components satisfy $\sigma_{n_I}^2 = \sigma_{n_Q}^2 = \sigma_n^2$.

Solution. Sketch of the standard argument: n_I and n_Q are obtained from n by the operations in §9.3, which are lowpass filters of one-sided width W applied to frequency-shifted copies. Their PSDs come out as $S_{n_I}(f) = S_{n_Q}(f) = S_N(f - f_c) + S_N(f + f_c)$ for $|f| < W$, i.e. flat of height N_0 over $[-W, W]$. So

$$\sigma_{n_I}^2 = \sigma_{n_Q}^2 = N_0 \cdot 2W = 2N_0 W = \sigma_n^2.$$

Equal variance to the bandpass noise itself — this is the fact that makes envelope-detector and coherent-detector SNR comparisons work out.

11 Problem set (do this last, timed)

Give yourself **60 minutes**. Solutions follow — do not look until you have finished.

1. Sketch $|X(f)|$ for $x(t) = \mathrm{rect}(t/T)\cos(2\pi f_c t)$ with $f_c \gg 1/T$. What is the bandwidth?

2. A real signal $x(t)$ has $X(f)$ with $X(f) = 0$ for $|f| > 5$ kHz. It is multiplied by $\cos(2\pi \cdot 100{,}000\, t)$. Over what frequency range does the result live? What is the minimum sampling rate needed to represent the *result* without loss if you sample it directly as a bandpass signal? (Hint: think about the lowpass equivalent.)

3. Prove that if $x(t)$ is real and even, $X(f)$ is real and even.

4. Compute $\left(e^{-at}u(t)\right) * \left(e^{-bt}u(t)\right)$ for $a \neq b$, two ways: directly, and via Fourier transforms with partial fractions.

5. A DSB-AM signal has $A = 1$, $\mu = 0.5$, and message $x(t) = \cos(2\pi \cdot 1000t)$. Compute the total power, the sideband power, and the power efficiency.

6. Find the Hilbert transform of $x(t) = \cos(2\pi f_0 t) + \sin(2\pi f_0 t)$, and hence write its analytic signal.

7. An LTI system has $h(t) = \delta(t) - \delta(t - T)$. Find $H(f)$, sketch $|H(f)|$, and state at which frequencies the system nulls the input. (This is a two-ray multipath channel — it shows up in the ISI chapter.)

8. $x(t)$ is bandlimited to $W = 4$ kHz and sampled at $f_s = 7$ kHz. If x contains a tone at 3.5 kHz, at what frequency does its alias appear after reconstruction with an ideal 3.5 kHz LPF?

Solutions

1. By the modulation property, $X(f) = \frac{T}{2}\operatorname{sinc}\left((f - f_c)T\right) + \frac{T}{2}\operatorname{sinc}\left((f + f_c)T\right)$: two sinc lobes centred at $\pm f_c$. First nulls at $f_c \pm 1/T$, so the null-to-null bandwidth is $2/T$. (The baseband pulse had bandwidth $1/T$; modulation doubles it. Every DSB scheme doubles bandwidth.)

2. The product occupies 95–105 kHz and -105 to -95 kHz: a bandpass signal with $f_c = 100$ kHz, $W = 5$ kHz, total bandwidth 10 kHz. Naive lowpass reasoning would demand $f_s > 2 \times 105 = 210$ kHz. But the lowpass equivalent x_ℓ is complex with bandwidth 5 kHz, so 10 kHz complex samples/second suffice — i.e. 10 kHz on each of the I and Q branches, 20 k real samples/second. This is *bandpass sampling*, and it is the practical reason §9 exists.

3. $X(f) = \int x(t)e^{-\mathrm{j}2\pi ft}\,\mathrm{d}t = \int x(t)\cos(2\pi ft)\,\mathrm{d}t - \mathrm{j}\int x(t)\sin(2\pi ft)\,\mathrm{d}t$. The second integrand is (even)$\times$(odd) $=$ odd, so it integrates to zero over $\mathbb{R}$. Hence $X(f)$ is real. And $X(-f) = \int x(t)\cos(2\pi ft)\,\mathrm{d}t = X(f)$ since cosine is even, so X is even. $\square$

4. Direct: for $t > 0$, $\int_0^t e^{-a\tau}e^{-b(t-\tau)}\,\mathrm{d}\tau = e^{-bt}\int_0^t e^{(b-a)\tau}\,\mathrm{d}\tau = \dfrac{e^{-at} - e^{-bt}}{b - a}$, and 0 for $t < 0$. Via FT: the product is $\dfrac{1}{(a + \mathrm{j}2\pi f)(b + \mathrm{j}2\pi f)} = \dfrac{1}{b - a}\left[\dfrac{1}{a + \mathrm{j}2\pi f} - \dfrac{1}{b + \mathrm{j}2\pi f}\right]$, which inverts term by term to the same answer. The second route is the one to use under time pressure.

5. $P_x = \frac{1}{2}$ for a unit-amplitude cosine. Total power $P = \frac{A^2}{2}(1 + \mu^2 P_x) = \frac{1}{2}(1 + 0.25 \cdot 0.5) = \frac{1}{2}(1.125) = 0.5625$ W. Carrier power $= \frac{A^2}{2} = 0.5$ W; sideband (message) power $= \frac{A^2\mu^2}{2}P_x = 0.0625$ W. Efficiency $\eta = 0.0625/0.5625 = 1/9 \approx 11.1\%$. Even at $\mu = 1$ with a sinusoidal message the ceiling is 33.3% — two-thirds of your transmitter power goes into a carrier that carries no information. That is the argument for DSB-SC.

6. $\hat{x}(t) = \sin(2\pi f_0 t) - \cos(2\pi f_0 t)$ (using $\widehat{\cos} = \sin$, $\widehat{\sin} = -\cos$). Then $x_+(t) = x + \mathrm{j}\hat{x} = (1 - \mathrm{j})\cos(2\pi f_0 t) + (1 + \mathrm{j})\sin(2\pi f_0 t)$. Alternatively note $x(t) = \sqrt{2}\cos(2\pi f_0 t - \pi/4)$, so $x_+(t) = \sqrt{2}\,e^{\mathrm{j}(2\pi f_0 t - \pi/4)}$ — much cleaner, and it confirms the analytic signal of a sinusoid is just the corresponding rotating phasor. (Convention note: with the $x_+ = \frac{1}{2}(x + \mathrm{j}\hat{x})$ definition used in §9, halve these.)

7. $H(f) = 1 - e^{-\mathrm{j}2\pi fT}$. Factor: $= e^{-\mathrm{j}\pi fT}\left(e^{\mathrm{j}\pi fT} - e^{-\mathrm{j}\pi fT}\right) = 2\mathrm{j}\sin(\pi fT)e^{-\mathrm{j}\pi fT}$, so $|H(f)| = 2|\sin(\pi fT)|$. Nulls where $\sin(\pi fT) = 0$, i.e. $f = k/T$ for integer k (including DC). A comb of nulls spaced $1/T$ apart — these are the "frequency-selective fades" of a two-path channel.

8. $f_s = 7$ kHz means replicas at $\pm 3.5 + 7k$. The tone at 3.5 kHz sits exactly at $f_s/2$: the replica from $k = -1$ lands at $3.5 - 7 = -3.5$ kHz, coinciding with the original negative-frequency line. This is the boundary case of the trap in §8.2 — the tone is not aliased to a different frequency, but its amplitude and phase are not recoverable in general (samples may all land on zero crossings). Note also that $f_s = 7 < 2W = 8$ kHz, so the signal is undersampled overall and other components between 3.5 and 4 kHz *are* genuinely aliased into the band.

12 One-page recall sheet

Transforms $X(f) = \int xe^{-j2\pi ft}\,dt$; $x(t) = \int Xe^{j2\pi ft}\,df$

Convolution $x * h \xleftrightarrow{\mathcal{F}} XH$ **Product** $xh \xleftrightarrow{\mathcal{F}} X * H$

Shift $x(t-t_0) \xleftrightarrow{\mathcal{F}} Xe^{-j2\pi ft_0}$ $xe^{j2\pi f_c t} \xleftrightarrow{\mathcal{F}} X(f-f_c)$

Modulation $x \cos 2\pi f_c t \xleftrightarrow{\mathcal{F}} \frac{1}{2}X(f-f_c) + \frac{1}{2}X(f+f_c)$

Scaling $x(at) \xleftrightarrow{\mathcal{F}} \frac{1}{|a|}X(f/a)$ **Duality** $X(t) \xleftrightarrow{\mathcal{F}} x(-f)$

Real $x \Rightarrow X(-f) = X^*(f)$

Parseval $\int |x|^2\,dt = \int |X|^2\,df$ $X(0) = \int x\,dt$, $x(0) = \int X\,df$

Pairs $\mathrm{rect}(t/T) \xleftrightarrow{\mathcal{F}} T\,\mathrm{sinc}(fT)$; $\mathrm{tri}(t/T) \xleftrightarrow{\mathcal{F}} T\,\mathrm{sinc}^2(fT)$; $\cos 2\pi f_c t \xleftrightarrow{\mathcal{F}} \frac{1}{2}[\delta(f-f_c) + \delta(f+f_c)]$; $\frac{1}{\pi t} \xleftrightarrow{\mathcal{F}} -j\,\mathrm{sgn}\,f$; $\sum_n \delta(t-nT_s) \xleftrightarrow{\mathcal{F}} f_s \sum_k \delta(f-kf_s)$

Sampling $X_s(f) = f_s \sum_k X(f-kf_s)$; need $f_s > 2W$

Bandpass $x_{bp} = \mathrm{Re}\{x_\ell e^{j2\pi f_c t}\} = x_I \cos 2\pi f_c t - x_Q \sin 2\pi f_c t$; $A = \sqrt{x_I^2 + x_Q^2}$

Coherent detection output $\propto \cos\phi$ — zero at $\phi = 90°$

DSB-SC $x_\ell = Ax(t)$ (real, sign-changing $\Rightarrow$ envelope detection fails)

DSB-AM $x_\ell = A(1+\mu x)$; envelope detection works iff $\mu|x| < 1$; $\eta = \frac{\mu^2 P_x}{1+\mu^2 P_x}$

Noise $S_Y = |H|^2 S_X$; white: $S_N = N_0/2$, $R_N = \frac{N_0}{2}\delta(\tau)$

The five mistakes that cost the most marks

1. Mixing f and ω conventions mid-problem (stray 2π's).
2. Forgetting the minus sign in $x_{bp} = x_I \cos - x_Q \sin$.
3. Calling the bandwidth of a real lowpass signal $2W$ when the course means W (or vice versa) — state your convention.
4. Applying the modulation property without checking $f_c > W$, so the shifted copies actually don't overlap.
5. Using $|X(f)|^2$ (energy spectral density) for a modulated carrier or noise, which is a *power* signal and needs PSD.

Built against the EE341 (Autumn 2026) course introduction and the handwritten lecture notes in your course folder, so the notation matches what Prof. Chaporkar uses. Reference texts if you want more depth: Madhow, Introduction to Communication Systems, *Ch. 2 (this is the best match for the lowpass-equivalent material in §9); Lathi, Ch. 3–4 for signals and Fourier; Haykin & Moher Ch. 2 for the bandpass/Hilbert treatment.*